Validation and Calibration of the Remote Consolidated Approach of Reporting on Food Security (rCARI) Methodology

Preliminary results from the Central African Republic (CAR) pilot study

Version 2.0

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Executive Summary

During the COVID-19 pandemic, a remote version of the Consolidated Approach for Reporting Food Insecurity (rCARI) was developed as an adaptation of the standard CARI methodology to report on food insecurity. The rCARI method differed from the standard CARI only in the economic vulnerability dimension. While the standard CARI used FES (food expenditure share) or ECMEN (economic capacity to meet essential needs) as an economic vulnerability indicator, rCARI used the household's income source and income change in place of FES because of the difficulty of collecting expenditure data over the phone. The rCARI methodology is has not yet been validated against the standard CARI.

Using a survey experiment methodology with stratified sample, this study aimed to evaluate and validate the rCARI methodology of collecting and reporting on food security against the standard CARI methodology. The sample frame for the study was the national population within which a random sample of households were selected in 2 out of the 20 prefectures (ADMIN1). A total of 2500 households were randomly selected and listed across 25 villages (primary sampling units-PSUs) and interviewed. The prefectures selected were done purposively based on mobile phone coverage, food security situation & accessibility (considering the security situation on the ground). Actual study respondents were randomly enlisted for the study through a random-walk methodology in each village (ADMIN5). Listed households were then randomly assigned to one of two study groups, the Face-to-Face (F2F) group, which received standard CARI modules through F2F interviews and remote group which received the remote modules through phone interviews (rCARI).

In CAR, one of the key challenges was the low phone access (approximately 75% of listed households did not have access to mobile phones). To address this, 300 low-budget phones were randomly distributed to households that were randomly assigned to receive phone interviews but did not have access to phones. The study interviewed a total of 1,245 household all together (969 (92%) F2F and 276 (27%) remote interviews respectively) out of a total of 2080 potential eligible households.

This report compares different versions of CARI computed using either remote or F2F modules. Three CARI version were calculated and compared with the standard CARI. These include: 1) CARI using FES from remote surveys, 2) CARI using income variable from F2F surveys, and 3) CARI using income sources and income change variable (the remote version of this indicator was implemented at the height of the COVID-19 pandemic and thus, for the purposes of this study will be referred to as standard rCARI).

The results show that;

- Households without mobile phones (usually excluded in mobile phone (CATI) surveys) are significantly more food insecure (by up to 35% more) regardless of the CARI calculation methodology used.
- The standard rCARI methodlogy (income choice and income change variable) significantly underestimates AFI compared to the conventional CARI methodlogy.

- The aggregated CARI results show that, replacing the FES with the income source & income change indicator significantly underestimates food insecurity (by up to 24% points).
- Households, generally tend to report better food consumption (better FCS) and worse coping (LCS & rCSI) over the phone than they report in F2F.
- The form (questionnaire) length does not have any statistical effect on AFI classification.
- Phone ownership has a significant influence in AFI classification (between 14% & 19% using FES, and 21% & 31% using income and income change indicators respectively). No effect is found for survey modality.

1 Introduction

Food security assessment and reporting are at the core of WFP's mandate of addressing hunger and food insecurity. Over time, different methodologies (e.g., IPC, CARI, Cadre Harmonise, etc.) have been developed and used to assess and report global food insecurity each differing in scope, and purpose. A commonality that cuts across all methods is the use of primary household data collected through face-to-face interviews/surveys. However, in recent times particularly with the outbreak of the Covid-19 pandemic and attendant restrictions on movement and physical contact, data collection through conventional face-to-face method has been adversely affected. This negatively impacted the ability to assess food security needs because of the data gap created. Developing methodologies to accurately measure and interpret food insecurity especially in emergencies where physical access is restricted thus, is of great importance for WFP and many stakeholders. This is particularly important during emergencies when rapid assessment is required for advocacy to save lives and livelihoods but physical access to the field is restricted either because of health threats or security considerations for enumerators.

To leverage the increased penetration of technology in operational areas of WFP during the Covid lockdowns, a remote version of the Consolidated Approach for Reporting Food Insecurity (CARI)- the so-called rCARI- was developed as an adaptation of the standard CARI methodology and administered remotely (over the phone). The methodology provided a stop-gap solution during the lockdown, however, questions about the validity of food security numbers it produces remain. For example, how are the remote modules comparable with already standardized face-to-face modules in capturing food insecurity. Although there exist validated studies on the use of face-to-face modules in assessing and reporting on food insecurity, remote methodology such as rCARI remains scant until now.

The potential advantages of the methodology abound for humanitarian work particularly for hard-to-reach areas. For example, the remote methodology has potential to save lives by providing real-time food insecurity data on the exposed population within the shortest possible time to inform policy. Furthermore, it can result in better use of scarce resources to achieve high end results by rolling out few targeted modules to assess food insecurity of a larger population.

The overall objective of this study was to assess the validity of the remote consolidated approach for reporting food insecurity - rCARI module(s) against conventional CARI modules for measuring and reporting food insecurity. The study compared indicators collected through phone surveys with conventional face-to-face survey, considering the latter as the standard methodology. Specifically, food Consumption Score (FCS), reduced Coping Strategy Index (rCSI), Livelihood Coping Strategy (LCS), Food Expenditure Share (FES), and income sources and income change have been compared over the two modalities. The Economic Capacity to Meet Essential Needs (ECMEN) indicator which is sometimes used to aggregate CARI will not be tested in this study, as it requires the standard long expenditure module which cannot be collected over phone. In addition, ECMEN calculation requires having an updated value of the minimum expenditure basket (MEB) of the

study population, which is not available for the study areas. In its stead, a shortened version of the standard expenditure module was be used to test the possibility of collecting FES data over the phone.

As CAR has one of the lowest mobile phone penetration rates globally (approximately 70% of the population do not own/use mobile phones), testing the comparability of data collection modalities (Face-to-Face (F2F) vs. Phone (Remote)) should be done within the wider consideration factors that could confound data collection using phones remotely. For example, other factors aside the modules should be considered while reading results of any remote survey.

1. Mobile phone ownership (mobile phone penetration rate in the study area) could likely exclude the most vulnerable population from the sample.
2. Mobile phone network coverage: Even in countries with high penetration rates, network coverage could be a problem, especially when electricity cuts are frequent and local infrastructure, such as communication masts, are not evenly distributed (World Bank, 2020).
3. Response rate: It is well documented that response rate in phone surveys is usually lower compared to F2F surveys (L'Engle et al., 2018; World Bank, 2020).

These factors may lead to skewed sample bias from phone surveys especially if the goal to report national representative results/outcomes. Nevertheless, there are several statistical weighting techniques that can adjust the results of phone surveys using baseline variables like using demographics characteristics of the most recent face-to-face surveys (Nagpal et al. 2021; Glazerman et al. 2020; Valliant et al. 2013; Elkasabi, 2015). This allows the results of such exercises to be computed and extrapolated to reflect some level of representativeness.

However, the current study may not be affected by these confounding issues since the main objective is not to report results at national level but to assess the effectiveness of remote modules to collect and report on food insecurity using a statistically comparable stratified sample. Thus, this current analysis does not employ weighting techniques to adjust for sampling skewness and response rate. Nonetheless, the study design will allow for controlling sample skewness due to mobile phone ownership, and the underlying background variables that are associated with mobile coverage, and non-response rate over the phone.

2 Study design & sample

The study was implemented as a survey experiment with stratified sample comparing the outcomes of two statistically comparable samples drawn from the same population.

Selection of study areas and participating households followed a stratified random sampling approach (stratified by the different Admin Areas-biggest to smallest but not representative at each admin area) based on 3-point criteria developed and agreed together with the WFP country office (CO) in Bangui. The criteria include, **1. Mobile phone coverage, 2. Food security situation, & 3. Accessibility (considering the security situation on the ground).** On the mobile phone coverage, a full list of areas with phone network was compiled through Orange/Moove/Telecel (the biggest mobile phone operators with nationwide coverage). Admin areas with high phone coverage were included in the sampling frame to ensure the biggest share of households in those areas will have phone coverage because of the remote interview part of the study.

The study design followed rigorous randomization procedures to assign potential study respondents to one of two study groups: F2F and Remote groups. The systematic steps in selecting the different administrative areas and study households with annotated explanation is presented in figure 1.

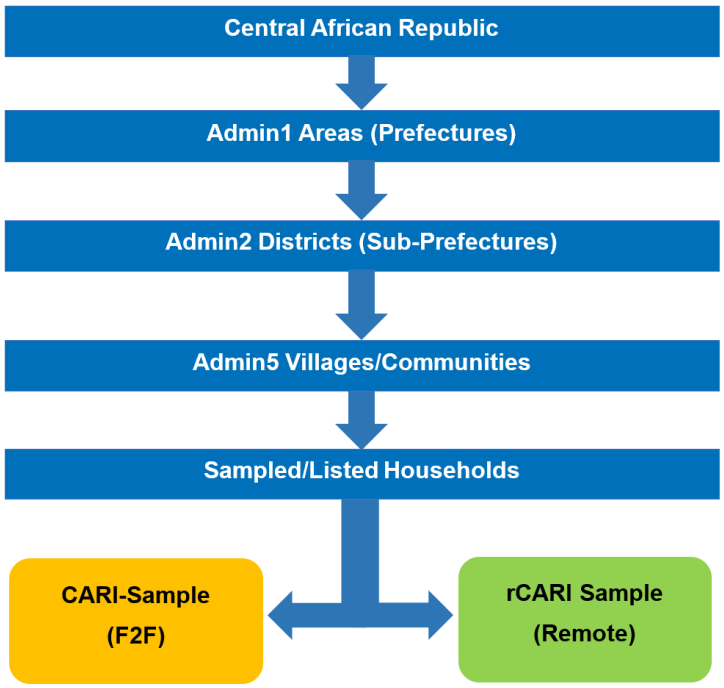

Figure 1: rCARI Validation Study Sampling Frame

1. STEP 1 - Based on the inclusion criteria (phone coverage, food security situation and accessibility), a list of prefectures meeting the compiled by the Country Office and two prefectures (Mbomou and Nana-Gribizi) were purposively selected as the study regions.

2. STEP 2 - Two sub-prefectures (ADMIN2), Bangassou and Kaga-Bandaro, were randomly selected from Mbomou and Nana-Gribizi Prefectures respectively. The Administrative structure of CAR allowed for inclusion of Admin3 areas (communes), therefore we randomly selected 4 communes from the 2 sub-prefectures. Bangassou, Nana, Botto, and Kanga-Bandoro were the 4 communes selected.

3. STEP 3 - Selection of villages (ADMIN5): A list of villages in the selected communes - Nana, Botto, Kaga-Bandaro, and Bangassou that met the inclusion criteria set out in STEP 1 were compiled at this stage as primary sampling units (PSUs). A total of 25 PSUs were randomly selected (distributed uniformly across the selected communes) and households for the study drawn from. The selected villages took into consideration the possible variations between rural and urban food security. Therefore, the final PSUs was made up of 15 urban and 10 rural villages respectively. See annex for the full list of administrative areas selected, number of households listed and their phone ownership.

4. STEP 4 - Enlisting households for the study: The sampling frame for study was all households in selected PSUs. A minimum of 100 households per PSU were selected through the random walk approach (listing every 3rd house in the sequence) by trained enumerators. Consent of participation was taken from all HHs before being included into the study sample (a total of 2,500 HHs). Enumerators were asked to report network coverage during listing and 420 households were dropped from the list because there was no network coverage at the time of listing (see annexes for the distribution of the 2,080 households by admin areas). The final listed households in this stage were used for the randomization process in the next step.

5. STEP 5 - Random assignment of household (hh): Consented final sampled households in STEP 4 are randomly assigned into one of two survey modalities/groups; F2F (CARI group) and Remote (rCARI group). The experiment was designed such that the full sample was divided equally between the two modalities.

6. STEP 6 - Mobile phone assignment to households without phones. Give the low mobile phone penetration in CAR (more than 70% households are without access to phones), low budget phones were for 300 households without phones but randomly assigned to receive phone interviews (detail description of this process is given in section 2.2).

   All interviews were conducted over the same time frame to ensure that households in the two modalities are reporting their food security situation over the same window.

2.1 Survey implementation

The final listed for the study before randomization was 2500 households. Socioeconomic and demographic characteristics (such as age, gender education status of the household head, household size, and ownership of common assets etc.) were collected during the listing phase. The contact details (address and phone number if any) of the listed households were also collected during the

listing.

For the data collection, enumerators were trained by in-person RAM facilitators from WPF-HQ with support from RBD (Dakar) and the VAM (vulnerability, assessment and mapping) team in CAR country office. Enumerator training was done in French over a 3-day period with one additional day used for field testing, adjusting, and adapting the final questionnaires (F2F & Remote).

Interviews (both F2F & Remote) were conducted in local languages of each locality. Enumerator were sent to localities where they could speak the local dialects.

2.2 Phone assignment for remote modality

In CAR, as discussed earlier, not only is the phone connectivity an issue, but phone ownership is generally low (only 27% own phones from the national statistics). This implies that, a larger proportion of the population i.e., a little over 70% of households (usually lower income earners and likely to be food insecure) are automatically excluded from phone survey sample frame.

The situation in CAR thus presented a unique opportunity to investigate deeper to understand the type of households/respondents are usually excluded in such exercises, their food security situation and how they cope with shocks or stress associated with food insecurity.

In this regard, a budget line was created to provide small hand-held phones to a sub-sample of the listed households (households with network coverage but without mobile phones but assigned randomly to the remote group). The budget was for 300 phones, however since the final sample without phones was more than budgeted, 300 households/respondents were randomly selected and given pre-connected phones with SIM cards (refer to Figure 1 for the different levels of randomization was implemented in the study and the corresponding sample size at each level).

Several strategies were put in place to ensure the phones distribution achieved high success. First, all mobile phones were fully charged and fixed with operational SIM cards before being distributed to selected households/respondents (to address electricity access in rural areas and operation challenges associated with the new phone respectively). Second, there was coordination between the field team distributing the phones and the team conducting the phone interviews in Bangui to ensure that calls were made to households/respondents immediately they received the mobile phones from the field team.

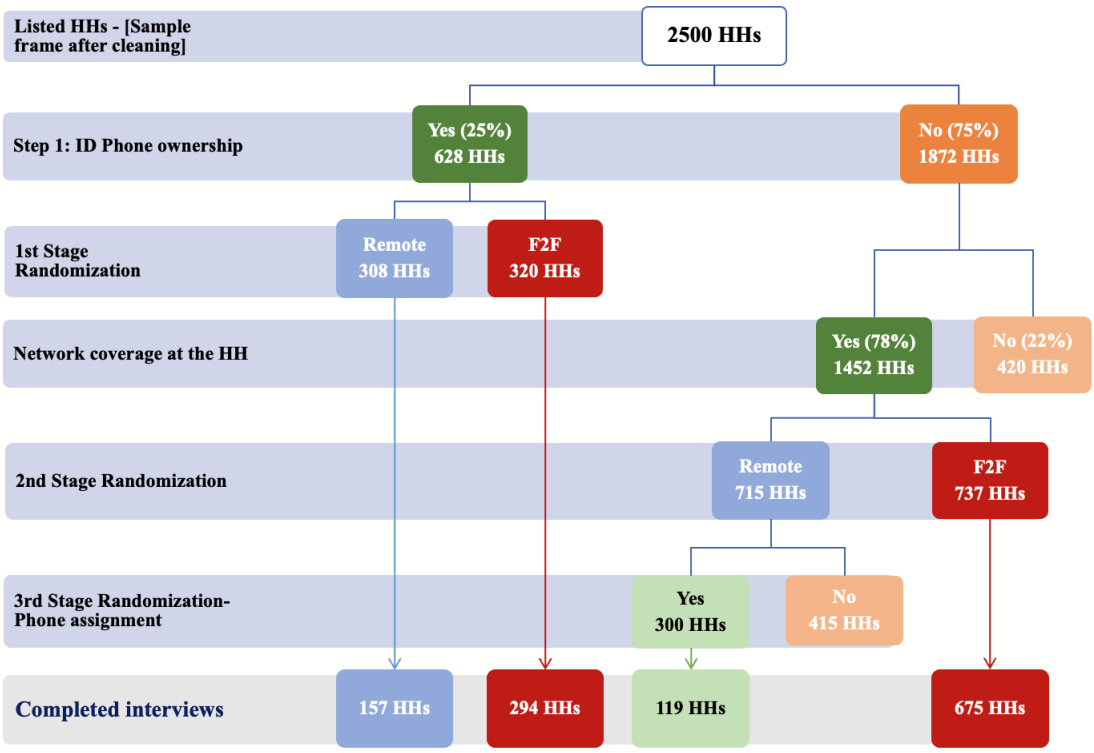

Figure 2: Sample Randomization and Survey Completion

Data collection was conducted from July - August 2023 in the selected study areas using trained enumerators from the statistical bureau of CAR-ICASEES. In the next section, the baseline comparison of the different groups is presented using the observable characteristics collected during the household listing phase.

3 Descriptive statistics

In this section, the core descriptive statistics of the study sample is presented. Given the rather elaborate design that has been implemented, a number of statistics in the form of interview completion, balance tables and attrition statistics across the different level of implementation are presented.

3.1 Survey completion

Table 2 shows the interview completion rate between F2F and Remote modalities. In total, the survey was able to complete 60% (1,245 interviews) of the original sample with the overall completion rate higher in the F2F (92%) compared to the Remote modality (27%).

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face-to-Face	88	8.33	969	91.67	1057	100.00
Remote	747	73.02	276	26.98	1023	100.00
Total	835	40.14	1245	59.86	2080	100.00

The study sample was sub-categorized into phone-owners and non-phone-owners based on the study design to capture phone and network coverage.

3.2 Randomization quality check

One of the cardinal requirements of the study design is the statistical comparability of the two study groups at the time of data collection (balance test) based on observable characteristics. To assess the statistical comparability (equivalence) of the two study groups - F2F and Remote - various baseline demographics, and other observable characteristics collected during the listing phase have been compared using the appropriate statistical tools. This helped to evaluate the equivalence of groups across the different levels of randomization implemented in the study. Specifically, the balance test was done for the following groups: 1) modality assignment (F2F & Remote), and 2) module length assignment (Long & Short).

3.2.1 Modality randomization

The statistical balance test for modality assignment is presented here. The sampled households have been categorized into tree different sub-samples and tested for their statistical comparability based on random assignment to the two study groups to receive F2F or Remote interviews. The three sub-sample are: 1) Phone-owners, 2) No phone sample, and 3) the full sample. Description of the different sub-samples is given in Figure 1.

The balance test for the modality assignment shows that, the randomization achieved a good statistical balance, i.e., households randomly assigned to the study groups were statistically simi-

lar/equivalent (using the information collected during the listing phase). The results are presented in Table 2.

Table 2: Balance Table for Randomized Survey Modality Assignment: Face-to-Face vs. Remote

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
List respondent age	35.97	37.59	1.61	39.42	39.35	-0.08	38.38	38.82	0.44
	(12.42)	(12.76)	(0.11)	(13.95)	(14.24)	(0.92)	(13.59)	(13.83)	(0.47)
Household head is the respondent	0.72	0.73	0.01	0.78	0.79	0.02	0.76	0.78	0.02
	(0.45)	(0.44)	(0.68)	(0.42)	(0.40)	(0.43)	(0.43)	(0.42)	(0.38)
Female Household Head	0.20	0.23	0.03	0.30	0.31	0.01	0.27	0.28	0.02
	(0.40)	(0.42)	(0.30)	(0.46)	(0.46)	(0.66)	(0.44)	(0.45)	(0.37)
Married Household Head	0.25	0.26	0.01	0.14	0.11	-0.03	0.17	0.16	-0.02
	(0.44)	(0.44)	(0.85)	(0.35)	(0.31)	(0.09)	(0.38)	(0.36)	(0.25)
Household head years of education	7.50	7.49	-0.02	3.39	3.30	-0.08	4.63	4.56	-0.07
	(4.10)	(4.50)	(0.96)	(3.27)	(3.22)	(0.63)	(4.01)	(4.12)	(0.70)
Displaced household	0.11	0.12	0.01	0.03	0.03	-0.00	0.06	0.06	0.00
	(0.31)	(0.32)	(0.67)	(0.18)	(0.18)	(0.85)	(0.23)	(0.23)	(0.86)
Household size	7.68	7.95	0.28	6.09	5.96	-0.13	6.57	6.56	-0.01
	(3.50)	(3.95)	(0.35)	(2.66)	(2.65)	(0.37)	(3.03)	(3.23)	(0.96)
Number of household member younger than 15	4.02	4.00	-0.02	3.21	3.13	-0.08	3.45	3.39	-0.06
	(2.27)	(2.12)	(0.93)	(1.92)	(1.90)	(0.41)	(2.06)	(2.01)	(0.47)
Number of household member older than 60	0.21	0.22	0.01	0.18	0.18	0.00	0.19	0.19	0.01
	(0.48)	(0.55)	(0.72)	(0.44)	(0.48)	(0.87)	(0.45)	(0.50)	(0.73)
Has household member with disability/chronic disease	0.29	0.30	0.00	0.15	0.16	0.01	0.19	0.20	0.01
	(0.46)	(0.46)	(0.96)	(0.36)	(0.37)	(0.50)	(0.40)	(0.40)	(0.59)
Number of rooms	2.84	2.99	0.15	1.96	1.91	-0.05	2.23	2.24	0.01
	(1.25)	(1.34)	(0.14)	(1.08)	(1.09)	(0.40)	(1.20)	(1.27)	(0.83)
Household Crowding Index	3.08	3.00	-0.08	4.02	4.00	-0.02	3.74	3.70	-0.03
	(1.79)	(1.64)	(0.57)	(2.94)	(2.97)	(0.92)	(2.68)	(2.68)	(0.78)
Household has impoverished roof	0.67	0.66	-0.01	0.85	0.86	0.01	0.79	0.80	0.00
	(0.47)	(0.48)	(0.80)	(0.36)	(0.35)	(0.57)	(0.41)	(0.40)	(0.79)
Household has impoverished floor	0.89	0.91	0.02	0.96	0.96	0.00	0.94	0.95	0.01
	(0.32)	(0.29)	(0.37)	(0.19)	(0.19)	(0.87)	(0.24)	(0.22)	(0.44)
Observations	320	308	628	737	715	1,452	1,057	1,023	2,080

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

This means that, the outcome indicator in this case, food security would likely be comparable between the groups holding all things constant and accounting for attrition.

3.2.2 Questionnaire length randomization-F2F

As described in the study set-up, aside from households being randomly assigned to received F2F and Remote, households were further randomized to receive either short- or long-modules of the FES, FCS and LCS. This section presents the balance test for this second level of randomization that was implemented in the study. Table 3 presents the balance test for the F2F group while Table 4 presents that of the Remote sub-group respectively. Both Table 3 & 4 show that statistically, household assignment to the short- and long-modules was successfully balanced.

3.3 Balance test for completed sample

Final interviews show a high number of attrition particularly among the remote survey sample. This could introduce an some level of bias especially if the attrition is systematic in nature. However, a balance test using only the completed sample show that despite the disproportionate attrition in F2F and remote samples, the final sample remained largely balanced with only minimal differences in magnitude (larger household size (0.8 people), higher education (1.4 years), and more number of room (0.4 room) in favour of the remote sample). See Table 5 for the full results.

Table 3: Balance Table for Randomized Questionnaire Length Assignment in F2F Survey: Long vs. Short

	Own Phone			No Phone			Total F2F		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
List respondent age	35.72	36.24	0.52	40.13	38.66	-1.47	38.81	37.92	-0.89
	(12.73)	(12.11)	(0.71)	(14.66)	(13.11)	(0.15)	(14.24)	(12.85)	(0.29)
Household head is the respondent	0.71	0.72	0.00	0.80	0.76	-0.04	0.77	0.75	-0.03
	(0.45)	(0.45)	(0.93)	(0.40)	(0.43)	(0.22)	(0.42)	(0.44)	(0.33)
Female Household Head	0.20	0.19	-0.01	0.30	0.30	-0.00	0.27	0.26	-0.01
	(0.40)	(0.40)	(0.84)	(0.46)	(0.46)	(0.94)	(0.44)	(0.44)	(0.85)
Married Household Head	0.24	0.26	0.02	0.12	0.16	0.03	0.16	0.19	0.03
	(0.43)	(0.44)	(0.70)	(0.33)	(0.37)	(0.18)	(0.37)	(0.39)	(0.19)
Household head years of education	7.30	7.71	0.41	3.32	3.46	0.13	4.52	4.76	0.24
	(4.04)	(4.16)	(0.38)	(3.23)	(3.32)	(0.58)	(3.94)	(4.09)	(0.34)
Displaced household	0.13	0.08	-0.06	0.04	0.02	-0.02	0.07	0.04	-0.03*
	(0.34)	(0.27)	(0.10)	(0.21)	(0.15)	(0.10)	(0.26)	(0.19)	(0.02)
Household size	7.46	7.91	0.45	6.17	6.00	-0.17	6.56	6.58	0.03
	(3.35)	(3.65)	(0.25)	(2.82)	(2.47)	(0.38)	(3.05)	(3.01)	(0.89)
Number of household member younger than 15	3.87	4.17	0.30	3.28	3.13	-0.15	3.46	3.45	-0.01
	(2.13)	(2.41)	(0.24)	(2.07)	(1.74)	(0.29)	(2.10)	(2.02)	(0.95)
Number of household member older than 60	0.21	0.21	0.00	0.19	0.16	-0.03	0.20	0.17	-0.02
	(0.49)	(0.47)	(0.94)	(0.43)	(0.45)	(0.30)	(0.45)	(0.45)	(0.43)
Has household member with disability/chronic disease	0.32	0.27	-0.05	0.16	0.14	-0.01	0.21	0.18	-0.02
	(0.47)	(0.44)	(0.35)	(0.36)	(0.35)	(0.61)	(0.40)	(0.39)	(0.34)
Number of rooms	2.78	2.90	0.12	2.02	1.89	-0.13	2.25	2.20	-0.05
	(1.25)	(1.24)	(0.38)	(1.10)	(1.05)	(0.10)	(1.20)	(1.20)	(0.50)
Household Crowding Index	3.04	3.13	0.08	3.88	4.17	0.29	3.63	3.85	0.22
	(1.71)	(1.89)	(0.67)	(2.76)	(3.13)	(0.18)	(2.52)	(2.85)	(0.18)
Household has impoverished roof	0.70	0.63	-0.06	0.84	0.86	0.02	0.79	0.79	-0.00
	(0.46)	(0.48)	(0.25)	(0.37)	(0.35)	(0.36)	(0.41)	(0.41)	(0.92)
Household has impoverished floor	0.92	0.85	-0.07	0.96	0.96	0.00	0.95	0.93	-0.02
	(0.27)	(0.36)	(0.06)	(0.19)	(0.19)	(0.85)	(0.22)	(0.26)	(0.19)
Observations	164	156	320	382	355	737	546	511	1,057

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 4: Balance Table for Randomized Questionnaire Length Assignment in Remote Survey: Long vs. Short

	Own Phone			No Phone			Total Remote		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
List respondent age	37.14	38.07	0.93	39.42	39.28	-0.14	38.71	38.93	0.22
	(13.19)	(12.29)	(0.52)	(14.01)	(14.48)	(0.90)	(13.79)	(13.88)	(0.80)
Household head is the respondent	0.75	0.71	-0.03	0.81	0.78	-0.03	0.79	0.76	-0.03
	(0.44)	(0.45)	(0.54)	(0.39)	(0.42)	(0.30)	(0.41)	(0.43)	(0.25)
Female Household Head	0.20	0.27	0.07	0.31	0.31	-0.00	0.28	0.29	0.02
	(0.40)	(0.44)	(0.17)	(0.46)	(0.46)	(0.89)	(0.45)	(0.46)	(0.52)
Married Household Head	0.23	0.29	0.06	0.11	0.11	-0.00	0.15	0.16	0.01
	(0.42)	(0.46)	(0.21)	(0.32)	(0.31)	(0.92)	(0.36)	(0.37)	(0.53)
Household head years of education	7.69	7.27	-0.42	3.51	3.10	-0.41	4.81	4.31	-0.49
	(4.35)	(4.66)	(0.41)	(3.31)	(3.13)	(0.09)	(4.14)	(4.10)	(0.06)
Displaced household	0.11	0.12	0.01	0.04	0.03	-0.01	0.06	0.05	-0.01
	(0.32)	(0.33)	(0.77)	(0.19)	(0.16)	(0.29)	(0.24)	(0.23)	(0.58)
Household size	7.99	7.92	-0.07	5.91	6.02	0.11	6.55	6.57	0.02
	(3.30)	(4.57)	(0.88)	(2.79)	(2.51)	(0.58)	(3.11)	(3.36)	(0.93)
Number of household member younger than 15	3.96	4.05	0.08	3.08	3.17	0.08	3.36	3.42	0.07
	(2.01)	(2.25)	(0.73)	(2.02)	(1.76)	(0.55)	(2.06)	(1.95)	(0.59)
Number of household member older than 60	0.24	0.20	-0.04	0.14	0.22	0.07*	0.17	0.21	0.04
	(0.60)	(0.48)	(0.54)	(0.40)	(0.55)	(0.04)	(0.47)	(0.53)	(0.22)
Has household member with disability/chronic disease	0.24	0.36	0.12*	0.18	0.15	-0.04	0.20	0.21	0.01
	(0.43)	(0.48)	(0.02)	(0.39)	(0.35)	(0.19)	(0.40)	(0.41)	(0.70)
Number of rooms	3.04	2.95	-0.09	1.89	1.94	0.06	2.24	2.23	-0.01
	(1.39)	(1.29)	(0.55)	(1.10)	(1.07)	(0.49)	(1.31)	(1.23)	(0.91)
Household Crowding Index	2.95	3.06	0.11	4.07	3.94	-0.13	3.72	3.68	-0.04
	(1.29)	(1.96)	(0.56)	(3.09)	(2.84)	(0.55)	(2.71)	(2.65)	(0.81)
Household has impoverished roof	0.66	0.65	-0.01	0.85	0.86	0.01	0.79	0.80	0.01
	(0.48)	(0.48)	(0.92)	(0.36)	(0.34)	(0.68)	(0.41)	(0.40)	(0.70)
Household has impoverished floor	0.94	0.88	-0.06	0.97	0.96	-0.01	0.96	0.94	-0.02
	(0.24)	(0.33)	(0.07)	(0.18)	(0.19)	(0.68)	(0.20)	(0.24)	(0.13)
Observations	161	147	308	358	357	715	519	504	1,023

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 5: Balance Table by Randomized Survey Modality: Face-to-Face vs. Remote for Completed Surveys

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
List respondent age	36.27	39.03	2.76*	39.47	39.76	0.28	38.50	39.34	0.84
	(12.55)	(12.38)	(0.03)	(13.82)	(13.94)	(0.84)	(13.52)	(13.05)	(0.35)
Household head is the respondent	0.72	0.78	0.05	0.79	0.82	0.04	0.77	0.80	0.03
	(0.45)	(0.42)	(0.21)	(0.41)	(0.38)	(0.34)	(0.42)	(0.40)	(0.29)
Female Household Head	0.19	0.25	0.06	0.30	0.32	0.02	0.26	0.28	0.02
	(0.39)	(0.44)	(0.12)	(0.46)	(0.47)	(0.62)	(0.44)	(0.45)	(0.55)
Married Household Head	0.27	0.27	0.01	0.14	0.14	-0.00	0.18	0.22	0.04
	(0.44)	(0.45)	(0.85)	(0.35)	(0.35)	(0.98)	(0.38)	(0.41)	(0.19)
Household head years of education	7.55	7.92	0.38	3.41	3.54	0.13	4.66	6.03	1.37***
	(4.13)	(4.69)	(0.40)	(3.25)	(3.49)	(0.70)	(4.02)	(4.73)	(0.00)
Displaced household	0.09	0.15	0.06	0.03	0.03	0.00	0.05	0.10	0.05**
	(0.29)	(0.36)	(0.07)	(0.17)	(0.18)	(0.89)	(0.22)	(0.30)	(0.01)
Household size	7.72	8.39	0.67*	6.08	5.99	-0.08	6.57	7.36	0.78***
	(3.44)	(3.40)	(0.05)	(2.67)	(3.05)	(0.78)	(3.02)	(3.46)	(0.00)
Number of household member younger than 15	4.04	4.31	0.27	3.23	3.13	-0.10	3.48	3.80	0.32*
	(2.27)	(2.03)	(0.20)	(1.95)	(2.29)	(0.64)	(2.08)	(2.22)	(0.03)
Number of household member older than 60	0.21	0.21	-0.00	0.17	0.13	-0.04	0.19	0.18	-0.01
	(0.49)	(0.44)	(0.99)	(0.44)	(0.41)	(0.33)	(0.46)	(0.43)	(0.78)
Has household member with disability/chronic disease	0.30	0.31	0.01	0.15	0.15	0.00	0.19	0.24	0.05
	(0.46)	(0.46)	(0.83)	(0.35)	(0.36)	(0.90)	(0.39)	(0.43)	(0.10)
Number of rooms	2.89	3.13	0.24	1.99	1.95	-0.04	2.26	2.62	0.36***
	(1.24)	(1.27)	(0.05)	(1.08)	(1.09)	(0.72)	(1.20)	(1.33)	(0.00)
Household Crowding Index	3.02	2.97	-0.05	3.93	3.80	-0.13	3.66	3.33	-0.33*
	(1.70)	(1.26)	(0.73)	(2.87)	(2.68)	(0.62)	(2.61)	(2.04)	(0.03)
Household has impoverished roof	0.68	0.63	-0.05	0.85	0.86	0.01	0.80	0.73	-0.07*
	(0.47)	(0.48)	(0.33)	(0.36)	(0.35)	(0.78)	(0.40)	(0.45)	(0.02)
Household has impoverished floor	0.90	0.92	0.02	0.96	0.97	0.01	0.94	0.94	-0.00
	(0.30)	(0.28)	(0.50)	(0.19)	(0.16)	(0.46)	(0.23)	(0.23)	(0.94)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * * p<0.05, ** p<0.01, *** p<0.001.

Given that results of the household randomization is balanced based on baseline observables, the main results of study showing the different outcome indicators calculated are presented next (for the results of the post survey attrition analysis, refer to the attrition tables in the Annexes).

4 Results & discussions

In this section, the main results of the study aggregated by the food security indicator are analysed and presented. Here, the different versions of CARI calculated using remote or F2F is compared with the standard CARI. A test of group means using student t-test of equivalence is performed. The comparison is at the sub-sample level of phone-ownership, no phone, and the combined full sample.

4.1 CARI vs. rCARI

Conventional/standard CARI: CARI-FES: Standard CARI using FES from F2F surveys.

Other ways of aggregating CARI (that are tested)

- rCARI-FES: Standard CARI using FES from remote surveys.
- CARI-Inc: CARI using the income variable from the F2F surveys.
- rCARI-Inc: Standard rCARI using income and income change variable from remote surveys (this is the CARI version that was used during COVID 19 Pandemic).

The different methods of calculating CARI that are tested against the standard/conventional CARI are shown in Table 6.

Table 6: The different versions of CARI & their indicators

	Aggregation Methodology			
Indicator	**CARI (F2F)**	**CARI (F2F)**	**rCARI (Remote)**	**rCARI (Remote)**
FCS	Yes	Yes	Yes	Yes
FES	Yes	**No**	Yes	**No**
LCS	Yes	Yes	Yes	Yes
rCSI	Yes	Yes	Yes	Yes
Income source & income change	**No**	Yes	**No**	Yes

For more information on the calculation of food security indicators included in each methodology refer to Annexes attached.

The first comparison is between CARI aggregated using remote modules and replacing FES with income source and income change variable (rCARI-Inc). The results are presented Table 7 is for all the 3 sub-sample comparisons.

The main significant finding across the results in Table 7 is that rCARI-Inc, significantly under-reports the number of food insecure people. The difference is found to be 21 percentage points amongst phone-owners sub-sample, 9 percentage points in the no-phone group and up to 23 percentage points in the combined full sample. This means that, the income source in its current form (as was used during the covid pandemic) likely underestimated the number of people that were food insecure compared to what would have been found if the standard CARI-FES was implemented around the same time.

Table 7: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.02	0.02	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.13)	(0.15)	(0.05)	(0.10)	(0.17)
Marginally Food Secure	0.31	0.52	0.21***	0.07	0.14	0.07*	0.14	0.36	0.22***
	(0.46)	(0.50)	(0.00)	(0.25)	(0.35)	(0.03)	(0.35)	(0.48)	(0.00)
Moderately Food Insecure	0.60	0.45	-0.15**	0.74	0.70	-0.04	0.70	0.56	-0.14***
	(0.49)	(0.50)	(0.00)	(0.44)	(0.46)	(0.39)	(0.46)	(0.50)	(0.00)
Severely Food Insecure	0.08	0.02	-0.06**	0.20	0.14	-0.05	0.16	0.07	-0.09***
	(0.27)	(0.14)	(0.00)	(0.40)	(0.35)	(0.14)	(0.37)	(0.26)	(0.00)
Food Insecure Group	0.68	0.47	-0.21***	0.93	0.84	-0.09**	0.86	0.63	-0.23***
	(0.47)	(0.50)	(0.00)	(0.25)	(0.37)	(0.01)	(0.35)	(0.48)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

While on the overage the remote modality slightly underestimates the number overall food insecure, across the sub-samples compared to F2F, comparing the non-phone owners with phone-owners, there are between 25 and 37 percent more food insecure people estimated in the F2F and Remote respectively in non-phone owners. Given that the research objective seeks to compare the results of the standard F2F with remote surveys, it is important to compare the results of the F2F results from the full sample with the remote results from the phone owners' sub-sample. The result of this comparison shows that using the remote methodology in its current setup using the income source and income change indicators significantly underreports acute food insecurity by 39 percentage points.

The next level of comparison is the CARI computed using FES from remote surveys with the standard CARI modules presented in Table 8. The results show different patterns across the 3 groups. First, for the phone-owners although the rCARI-FES in terms of absolute terms produces higher number of food insecure people, the difference is not statistically significantly compared to the standard CARI-FES. Second, for the no phone owners and the full sample, rCARI-FES significantly under reports the food insecure numbers by between 7 and 8 percentage points. The overall effect is driven by the moderately food insecure category where rCARI-FES significantly under reports the numbers by 15 and 11 percentage points respectively for non-phone owners and full-sample.

The results of the comparison in Table 8 shows that using the FES module in phone surveys will likely result in under-reporting food insecurity compared to the standard F2F FES module for the full sample.

A cursory analysis of the results in Table 8 show that (as expected) the non-phone owners are significantly more food insecure (moderately food insecure plus severely food insecure groups combined). The FES indicator reports slightly higher number of food insecure people 68 percent (phone owners), 93 percent (non-phone owners) in the F2F modality while reporting 71 percent (phone owners) and 87 percent(non-phone owners) in the remote modality. While the F2F seem to produce lower AFI among the phone owners, among the phone owners the reverse is true with the remote

Table 8: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.01	0.01	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.09)	(0.32)	(0.05)	(0.08)	(0.33)
Marginally Food Secure	0.31	0.29	-0.02	0.07	0.13	0.06	0.14	0.22	0.08**
	(0.46)	(0.45)	(0.61)	(0.25)	(0.33)	(0.07)	(0.35)	(0.41)	(0.01)
Moderately Food Insecure	0.60	0.58	-0.02	0.74	0.59	-0.15**	0.70	0.58	-0.11***
	(0.49)	(0.50)	(0.65)	(0.44)	(0.49)	(0.00)	(0.46)	(0.49)	(0.00)
Severely Food Insecure	0.08	0.13	0.05	0.20	0.28	0.08	0.16	0.19	0.03
	(0.27)	(0.33)	(0.14)	(0.40)	(0.45)	(0.06)	(0.37)	(0.39)	(0.24)
Food Insecure Group	0.68	0.71	0.02	0.93	0.87	-0.07*	0.86	0.78	-0.08**
	(0.47)	(0.46)	(0.61)	(0.25)	(0.34)	(0.04)	(0.35)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

producing relatively lower AFI numbers.

In the next analysis, CARI computed using income source and income change variable is compared over the modality. The same module asked over phone is compared with the F2F responses. The result of this comparison is presented in Table 9.

The results show that, amongst the phone-owners sample, there is no statistical difference in the food insecurity outcomes between the remote and F2F. However, amongst the non-phone owners' group, the income change indicator significantly over states the number of severely food insecure by 11 percentage points over the phone (classifying the overall food insecure by 8 percentage points) compared to the F2F module. The results looking at the combined sample show that though the remote version shows a bi-directional effect underestimating the number of moderately food insecure (8%) and overestimating the severely food insecure (4%).

Table 9: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.02	0.01	-0.01	0.00	0.02	0.02	0.01	0.01	0.00
	(0.14)	(0.08)	(0.18)	(0.04)	(0.13)	(0.20)	(0.08)	(0.10)	(0.59)
Marginally Food Secure	0.51	0.52	0.01	0.24	0.14	-0.09**	0.32	0.36	0.04
	(0.50)	(0.50)	(0.86)	(0.43)	(0.35)	(0.01)	(0.47)	(0.48)	(0.25)
Moderately Food Insecure	0.45	0.45	0.01	0.72	0.70	-0.03	0.64	0.56	-0.08*
	(0.50)	(0.50)	(0.89)	(0.45)	(0.46)	(0.55)	(0.48)	(0.50)	(0.02)
Severely Food Insecure	0.02	0.02	-0.00	0.04	0.14	0.11**	0.03	0.07	0.04*
	(0.14)	(0.14)	(0.92)	(0.19)	(0.35)	(0.00)	(0.18)	(0.26)	(0.01)
Food Insecure Group	0.47	0.47	0.01	0.76	0.84	0.08*	0.67	0.63	-0.04
	(0.50)	(0.50)	(0.91)	(0.43)	(0.37)	(0.04)	(0.47)	(0.48)	(0.21)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

The next level of analysis looks at the food security outcomes using a paired sample t-test of means. Here, CARI is calculated for the same household/respondent using FES and Income source and income change separately. Table 10 presents the comparison for the paired F2F sample while Table 11 presents that for the remote sample.

Table 10 show that the income indicator significantly under-reports food insecurity by between 17

and 22 percentage points across all sub-samples. In terms of the non-phone owners and phone owners, the difference is much pronounced (about 30 percentage more using the income indicator and 25 percentage points using the FES: See Table 11).

Table 10: Impact of Indicator Composition: Paired CARI-FES vs. CARI-Inc in F2F Surveys

	Own Phone			No Phone			Total		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food Secure	0.01	0.02	0.01**	0.00	0.00	0.00	0.00	0.01	0.01*
	(0.00)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Marginally Food Secure	0.31	0.51	0.2***	0.07	0.24	0.17***	0.14	0.32	0.18***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)
Moderately Food Insecure	0.60	0.45	-0.16***	0.74	0.72	-0.01	0.70	0.64	-0.06**
	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.01)	(0.02)	(0.02)
Severely Food Insecure	0.08	0.02	-0.06***	0.20	0.04	-0.16***	0.16	0.03	-0.13***
	(0.02)	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)
Food Insecure Group	0.68	0.47	-0.22***	0.93	0.76	-0.17***	0.86	0.67	-0.18***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)
Observations	294	294	294	675	675	675	969	969	969

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

In Table 11, the results still show that if the income indicator was used to calculate food security, households/respondents tend to significantly underreport their food insecurity (24% for phone owners & 14% for the combined sample). Although the difference is significantly different amongst the non-phone owners' sub-sample, it can be observed that using the income indicator respondents regardless of the modality to under-report their food insecurity compared to using FES module.

Table 11: Impact of Indicator Composition: Paired CARI-FES vs. CARI-Inc in Remote Surveys

	Own Phone			No Phone			Total		
Food Security Category	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff
Food Secure	0.01	0.01	0.00	0.01	0.02	0.01	0.01	0.01	0.00
	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)
Marginally Food Secure	0.29	0.52	0.24***	0.13	0.14	0.02	0.22	0.36	0.14***
	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)
Moderately Food Insecure	0.58	0.45	-0.13*	0.59	0.70	0.11*	0.58	0.56	-0.03
	(0.04)	(0.04)	(0.05)	(0.05)	(0.04)	(0.05)	(0.03)	(0.03)	(0.04)
Severely Food Insecure	0.13	0.02	-0.11***	0.28	0.14	-0.13**	0.19	0.07	-0.12***
	(0.03)	(0.01)	(0.03)	(0.04)	(0.03)	(0.04)	(0.02)	(0.02)	(0.02)
Food Insecure Group	0.71	0.47	-0.24***	0.87	0.84	-0.03	0.78	0.63	-0.14***
	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Observations	157	157	157	119	119	119	276	276	276

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

To conclude on the CARI and rCARI comparisons, three important highlights are observed. First, the sample that did not own phones (non-phone owners sub-sample) are consistently more food insecure regardless of which modality/module is used to collect the data (76% food insecure in F2F & 84% in remote modality). Thus, in a typical CATI (computer assisted telephone interviews) surveys using RDD (random digit dialing) where only a sub-population of respondents with access to phones are interviewed, it automatically exclude non-phone owners who are likely to be more food insecure and vulnerable. Second, the remote CARI methodology regardless of the indicators used to compute AFI significantly under-reports AFI. This is found to be between 15 and 39 percentage points. This means that people tend to generally under-report their food insecurity over the phone. Third, using the income source and income change indicator instead of the FES significantly over reports food security regardless of the modality (F2F or remote) used in collecting it.

4.2 Comparison of the food security indicators across modality

In this section, the results of the individual indicators compared over the two modalities (F2F vs remote) are compared statistically and presented.

4.2.1 Food Consumption Score (FCS)

The Food Consumption Score (FCS) is a composite score based on households' dietary diversity, food consumption frequency, and relative nutritional importance of different food groups, and is calculated by inspecting how often households consume food items from the different food groups during a 7-day reference period. The responses are aggregated, and weights applied to obtain an index which is then used to categorize the household into one of three groups of food consumption: food secure, borderline, and poor.

The results show that consistently, the remote FCS module significantly overreported food security (reporting better food consumption) compared to the F2F module; 8 points among the phone owners' sub-sample, 4 & 10 points more among the non-phone owners and the full sample respectively (see Table 12). These difference between the F2F, and remote responses is driven by higher consumption of tubers and cereals, protein-rich foods, dairy products, vegetables, and sugar over the phone.

In terms of the three FCS Groups, consistently, the remote module under reports the number of households with poor food consumption by between 12 and 23 percentage points.

Table 12: Impact of Survey Modality on Food Consumption Score and Groups

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Consumption Score	39.76	47.63	7.87***	24.59	28.12	3.53**	29.19	39.22	10.03***
	(17.58)	(17.45)	(0.00)	(12.67)	(12.00)	(0.00)	(15.94)	(18.12)	(0.00)
FCS: Acceptable	0.40	0.58	0.18***	0.10	0.09	-0.01	0.19	0.37	0.18***
	(0.49)	(0.50)	(0.00)	(0.31)	(0.29)	(0.70)	(0.40)	(0.48)	(0.00)
FCS: Borderline	0.35	0.29	-0.06	0.24	0.39	0.15**	0.27	0.33	0.06
	(0.48)	(0.45)	(0.19)	(0.43)	(0.49)	(0.00)	(0.45)	(0.47)	(0.07)
FCS: Poor	0.25	0.13	-0.12**	0.66	0.52	-0.14**	0.53	0.30	-0.23***
	(0.43)	(0.34)	(0.00)	(0.48)	(0.50)	(0.01)	(0.50)	(0.46)	(0.00)
Consumption over the past 7 days (cereals and tubers)	5.87	6.24	0.36*	5.08	4.87	-0.21	5.32	5.64	0.33*
	(1.86)	(1.33)	(0.02)	(2.17)	(2.13)	(0.31)	(2.11)	(1.85)	(0.01)
Consumption over the past 7 days (pulses)	3.74	3.74	-0.01	2.01	1.98	-0.03	2.54	2.98	0.45**
	(2.33)	(2.08)	(0.98)	(1.92)	(1.62)	(0.88)	(2.20)	(2.08)	(0.00)
Consumption over the past 7 days (dairy products)	0.50	1.14	0.64**	0.07	0.40	0.33**	0.20	0.82	0.62***
	(1.57)	(2.27)	(0.00)	(0.55)	(1.11)	(0.00)	(0.99)	(1.90)	(0.00)
Consumption over the past 7 days (protein-rich foods)	1.72	2.09	0.37*	0.51	0.59	0.08	0.88	1.44	0.56***
	(1.81)	(1.81)	(0.04)	(1.07)	(0.99)	(0.45)	(1.45)	(1.68)	(0.00)
Consumption over the past 7 days (vegetables)	3.73	5.64	1.92***	4.11	5.56	1.46***	3.99	5.61	1.62***
	(2.93)	(1.66)	(0.00)	(2.75)	(1.78)	(0.00)	(2.81)	(1.71)	(0.00)
Consumption over the past 7 days (fruit)	0.54	0.72	0.18	0.31	0.59	0.27*	0.38	0.66	0.28**
	(1.31)	(1.34)	(0.17)	(1.03)	(1.22)	(0.02)	(1.13)	(1.29)	(0.00)
Consumption over the past 7 days (oil)	4.90	5.79	0.89***	1.99	2.78	0.79***	2.87	4.49	1.62***
	(2.37)	(1.81)	(0.00)	(2.38)	(2.24)	(0.00)	(2.73)	(2.50)	(0.00)
Consumption over the past 7 days (sugar)	2.36	3.53	1.17***	1.30	1.86	0.55**	1.62	2.81	1.18***
	(2.60)	(2.78)	(0.00)	(2.02)	(1.98)	(0.01)	(2.26)	(2.60)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.2 Reduced Coping Strategies Index (rCSI)

The reduced Coping Strategies index (rCSI) is used to compare households' stress levels due to food shortages. It measures the frequency and severity of the food consumption related behaviours by households to cope with food shortage within a 7day recall period. The indicator is comprised of 5 standard strategies framed around how and who consumes food in times of shortage. It serves as a proxy indicator for food access.

Table 13 presents the results of the rCSI module for the three sub-samples. The results show that consistently, households reported higher rCSI (2.9 points amongst the phone owners' group, 3.95 in the non-phone owners and 3.52 in the combined sample respectively). This trend is further reflected in classifying more respondents as being food secure (rCSI severe category). The difference is mainly driven by more households over the phone restricting adult food consumption, reducing meals per day and meal sizes eaten.

Table 13: Impact of Survey Modality on Reduced Coping Strategy Index Classification

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Reduced Coping Strategies Index (rCSI)	14.35	17.27	2.92**	13.16	16.75	3.59***	13.52	17.05	3.52***
	(9.97)	(8.88)	(0.00)	(10.70)	(9.82)	(0.00)	(10.49)	(9.28)	(0.00)
rCSI: Low	0.06	0.04	-0.02	0.19	0.07	-0.13***	0.15	0.05	-0.10***
	(0.25)	(0.21)	(0.36)	(0.40)	(0.25)	(0.00)	(0.36)	(0.23)	(0.00)
rCSI: Median	0.70	0.57	-0.14**	0.55	0.57	0.02	0.60	0.57	-0.03
	(0.46)	(0.50)	(0.00)	(0.50)	(0.50)	(0.66)	(0.49)	(0.50)	(0.41)
rCSI: Severe	0.23	0.39	0.16***	0.26	0.36	0.11*	0.25	0.38	0.13***
	(0.42)	(0.49)	(0.00)	(0.44)	(0.48)	(0.03)	(0.43)	(0.49)	(0.00)
Relied on less preferred, less expensive food	3.92	2.75	-1.17***	3.39	3.29	-0.10	3.55	2.98	-0.57***
	(2.00)	(1.84)	(0.00)	(2.43)	(2.45)	(0.67)	(2.32)	(2.14)	(0.00)
Borrowed food or relied on help from friends or relatives	0.77	0.89	0.12	0.82	0.74	-0.08	0.81	0.82	0.02
	(1.15)	(1.25)	(0.32)	(1.26)	(1.03)	(0.43)	(1.23)	(1.16)	(0.84)
Reduced portion size of meals at meals time	1.98	2.65	0.67**	2.01	2.45	0.44*	2.00	2.57	0.57***
	(2.27)	(2.14)	(0.00)	(2.39)	(2.09)	(0.04)	(2.35)	(2.12)	(0.00)
Restricted consumption by adults in order for young-children to eat	1.73	2.70	0.97***	1.52	2.42	0.90***	1.58	2.58	1.00***
	(1.90)	(1.83)	(0.00)	(1.80)	(2.16)	(0.00)	(1.83)	(1.98)	(0.00)
Reduced the number of meals eaten per day	1.74	2.00	0.26	1.55	2.27	0.72***	1.61	2.12	0.51***
	(2.38)	(2.12)	(0.24)	(2.42)	(2.06)	(0.00)	(2.41)	(2.09)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.3 Livelihood Coping Strategies (LCS)

The livelihood coping strategy indicator is a household-level indicator that measures the household's capacity to deal with the insufficient food consumption because of shortages or the lack of money in the medium- to long-term bases (30days to 12months) to meet their dietary requirements. Broadly, the strategies are categorized into 3 broad groups: stress, crisis, and emergency level. For purposes of targeting and assistance, only the LCS crisis and emergency categories are usually considered (these categories classify households into food insecure).

Table 14 shows the aggregated LCS results categorized into the 4 main categories. The results show some consistency in both magnitude and direction of the difference between F2F and remote, the remote sample consistently reported doing worse compared to the F2F sample (28% more households in the phone-owners sub-sample, 30% in the non-phone owners and 29% in the combined sample cumulatively for crisis and emergency categories). In general, reporting emergency coping was 4 to 5 times higher in the remote compared to F2F across the different sub-groups of analysis.

Table 14: Impact of Survey Modality on Livelihood Coping Strategy

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Maximum Livelihood Coping: None	0.20	0.11	-0.09*	0.39	0.13	-0.26***	0.33	0.12	-0.21***
	(0.40)	(0.32)	(0.01)	(0.49)	(0.34)	(0.00)	(0.47)	(0.33)	(0.00)
Maximum Livelihood Coping: Stress level	0.61	0.43	-0.19***	0.43	0.39	-0.04	0.48	0.41	-0.07*
	(0.49)	(0.50)	(0.00)	(0.50)	(0.49)	(0.39)	(0.50)	(0.49)	(0.03)
Maximum Livelihood Coping: Crisis level	0.14	0.26	0.13**	0.13	0.21	0.08*	0.13	0.24	0.11***
	(0.34)	(0.44)	(0.00)	(0.33)	(0.41)	(0.04)	(0.34)	(0.43)	(0.00)
Maximum Livelihood Coping: Emergency level	0.05	0.20	0.15***	0.05	0.27	0.22***	0.05	0.23	0.18***
	(0.22)	(0.40)	(0.00)	(0.22)	(0.45)	(0.00)	(0.22)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

To understand the composition of the different categories reported in Table 14, the individual strategies adopted are disaggregated and reported in Table 15. Among the phone-owners and non-phone owners' sub-samples, strategies such as selling productive assets, scavenging, reduced health care expenditure and selling places of dwelling drive the significant difference between the remote and F2F modality. However, in the full sample, the remote modality significantly adopted more of each strategy than the F2F modality.

Table 15: CAR: Each Coping Strategy Used

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Used Stress coping: Sold household assets/goods	0.28	0.46	0.18***	0.15	0.29	0.14**	0.19	0.38	0.20***
	(0.45)	(0.50)	(0.00)	(0.35)	(0.45)	(0.00)	(0.39)	(0.49)	(0.00)
Used Stress coping: Reduced expenses on health and education	0.23	0.43	0.20***	0.18	0.40	0.23***	0.19	0.42	0.22***
	(0.42)	(0.50)	(0.00)	(0.38)	(0.49)	(0.00)	(0.39)	(0.49)	(0.00)
Used Stress coping: Spent savings	0.47	0.50	0.03	0.22	0.31	0.09	0.30	0.42	0.12***
	(0.50)	(0.50)	(0.50)	(0.42)	(0.46)	(0.06)	(0.46)	(0.49)	(0.00)
Used Stress coping: Borrowed money	0.46	0.53	0.07	0.31	0.34	0.02	0.36	0.45	0.09**
	(0.50)	(0.50)	(0.18)	(0.46)	(0.47)	(0.62)	(0.48)	(0.50)	(0.01)
Used Crisis coping: Sold productive assets or means of transport	0.07	0.26	0.19***	0.02	0.07	0.05*	0.04	0.18	0.14***
	(0.26)	(0.44)	(0.00)	(0.14)	(0.25)	(0.05)	(0.19)	(0.38)	(0.00)
Used Crisis coping: Moved to somewhere else	0.05	0.14	0.09**	0.11	0.24	0.13**	0.09	0.18	0.09***
	(0.23)	(0.35)	(0.01)	(0.31)	(0.43)	(0.00)	(0.29)	(0.39)	(0.00)
Used Crisis coping: Withdrew children from school	0.03	0.05	0.02	0.03	0.09	0.06*	0.03	0.07	0.04*
	(0.17)	(0.22)	(0.32)	(0.17)	(0.29)	(0.02)	(0.17)	(0.25)	(0.01)
Used Emergency coping: Sold houses	0.03	0.14	0.11***	0.02	0.22	0.20***	0.02	0.17	0.15***
	(0.17)	(0.35)	(0.00)	(0.15)	(0.41)	(0.00)	(0.16)	(0.38)	(0.00)
Used Emergency coping: Begged and/or scavenged	0.01	0.07	0.06**	0.02	0.08	0.05*	0.02	0.07	0.05**
	(0.12)	(0.26)	(0.01)	(0.15)	(0.27)	(0.04)	(0.14)	(0.26)	(0.00)
Used Emergency coping: Engaged in illegal income activities	0.01	0.03	0.02	0.01	0.05	0.04*	0.01	0.04	0.03*
	(0.10)	(0.16)	(0.27)	(0.09)	(0.22)	(0.04)	(0.10)	(0.19)	(0.02)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.4 Food Expenditure Share (FES)

The food expenditure share (FES) is a proxy food security indicator used to measure the economic vulnerability of a household. It is given as the share of household's total consumption expenditure spent on food. The higher the share of food expenditure of a household, the more vulnerable/food insecure the household.

Table 16 presents the summarize FES variable across the different groups. The aggregate FES is ranging from 67 to 73 percentage depending on the whether the respondent is coming from the phone-owners' or non-phone owners' sub-samples. For the phone owners' sample, the remote modality estimates FES at 73 percentage (6 percentage points higher than F2F). The effect is

that, it significantly overestimates the number of households in category 4 (>= 75%) severely food insecure by 10 percentage points. On the non-phone owners' sub-sample, the aggregate FES is significantly lower in remote modality compared to the F2F modality by 6 percentage points reflected in the FES category 4 (underestimating this category by 11 percentage points).

Once the two samples are combined, there is no significant difference between the aggregate FES from remote and F2F modalities.

Table 16: Impact of Survey Modality on Expenditure and Food Expenditure Share

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Expenditure Share	0.67	0.73	0.06*	0.73	0.68	-0.06*	0.71	0.70	-0.01
	(0.20)	(0.24)	(0.01)	(0.17)	(0.25)	(0.02)	(0.18)	(0.24)	(0.52)
FES < 50 percent	0.20	0.16	-0.04	0.10	0.25	0.15***	0.13	0.20	0.07*
	(0.40)	(0.37)	(0.27)	(0.30)	(0.44)	(0.00)	(0.34)	(0.40)	(0.01)
FES: 50-65 percent	0.23	0.17	-0.06	0.16	0.18	0.02	0.18	0.17	-0.00
	(0.42)	(0.38)	(0.15)	(0.36)	(0.38)	(0.61)	(0.38)	(0.38)	(0.86)
FES: 65-75 percent	0.13	0.13	-0.01	0.17	0.11	-0.06	0.16	0.12	-0.04
	(0.34)	(0.33)	(0.87)	(0.38)	(0.31)	(0.06)	(0.37)	(0.33)	(0.08)
FES: >= 75 percent	0.44	0.54	0.10*	0.57	0.46	-0.11*	0.53	0.51	-0.02
	(0.50)	(0.50)	(0.04)	(0.50)	(0.50)	(0.03)	(0.50)	(0.50)	(0.50)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.5 Income Source and Income Change

The income indicator captures the respondent/household income source and income change experienced in the last 12 months prior to the interview. This indicator is a recent addition was introduced to calculate CARI in remote assessment during COVID-19 and it is not yet part of the to the CARI console. The indicator to reflects the economic situation of the respondent/household in the absence of the FES. This was especially important during the COVID-19 outbreak when exigencies of the time did not allow for the collection of expenditure to calculate FES. The type of income generating activity and how economic situation at the time impacted on income flow either positive or negative was considered a good proxy for vulnerable the household could be. Based on the type of income (regular/irregular) and the change to income (change/no change) to the income stream, the respondent/household is classified into one of 4 CARI categories (see Table 17 for the categories). The irregular & reduced, and no income sources groups corresponding to FES category 3 (FES: 65-75%) and category 4 (FES: >= 75%), respectively.

The results of the income source and income change indicator comparison over modality shows that, respondents tend to underreport their food security over the phone (more households reported stable regular income source in the F2F interviews than remote across all the sub-samples). Although this may be driven by the categorization of agricultural income as regular income, the effect should be same across modalities. On the other hand, respondents reported engaging in significantly more irregular income sources (unstable and reducing income) over the phone compared to F2F.

Table 17: Impact of Survey Modality on Income

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Income: Regular and Unchanged/Increasing	0.45	0.26	-0.19***	0.44	0.24	-0.19***	0.44	0.25	-0.19***
	(0.50)	(0.44)	(0.00)	(0.50)	(0.43)	(0.00)	(0.50)	(0.44)	(0.00)
Income: Regular and Reduced OR Irregular and Unchanged/Increasing	0.42	0.50	0.08	0.46	0.49	0.02	0.45	0.49	0.04
	(0.49)	(0.50)	(0.13)	(0.50)	(0.50)	(0.63)	(0.50)	(0.50)	(0.22)
Income: Irregular and Reduced	0.12	0.23	0.11**	0.08	0.25	0.17***	0.09	0.24	0.15***
	(0.32)	(0.42)	(0.00)	(0.27)	(0.44)	(0.00)	(0.29)	(0.43)	(0.00)
Income: No income sources	0.01	0.01	-0.00	0.02	0.02	-0.00	0.02	0.01	-0.00
	(0.12)	(0.11)	(0.94)	(0.14)	(0.13)	(0.85)	(0.13)	(0.12)	(0.71)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.3 CARI using FES vs CARI Income source & income change

The results comparing CARI to different version of rCARI so far indicates that, the two main drivers of the difference by replacing FES with the income indicator. In this section, a simple sensitivity analysis is performed on these two indicators to see how the income indicator performs compared with FES.

The cross tabulation presented show that, there is a weak correlation between the two indicators with more of the divergence occurring as the two indicators move towards food insecure, i.e., CARI categories 3 & 4 (see Table 18). In classifying the food insecurity, the income source and income change indicator correctly classify 69 percent of the households classified by FES as moderately food insecure and only 17 percent of the households classified as severely food insecure.

Table 18: Cross Tabulation of CARI_FES vs. CARI_Income

	FES									
	Food Secure		Marginally Food Secure		Moderately Food Insecure		Severely Food Insecure		Total	
	Count	Percent	Count	Percent	Count	Percent	Count	Percent	Count	Percent
Food Secure	4	100.00	0	0.00	0	0.00	0	0.00	4	100.00
Marginally Food Secure	6	3.05	171	86.80	20	10.15	0	0.00	197	100.00
Moderately Food Insecure	0	0.00	239	28.62	580	69.46	16	1.92	835	100.00
Severely Food Insecure	0	0.00	0	0.00	174	83.25	35	16.75	209	100.00
Total	10	0.80	410	32.93	774	62.17	51	4.10	1245	100.00

From Table 18, the following percentages can be calculated, and conclusions derived.

- Good match = 15% (this is the percentage of households that have been correctly predicted by both indicators as being in the same food security category).
- Close match upper bound = 4% (this is the percentage of households representing that have been wrongly classified into a higher food security category by the CARI using the income indicator, e.g., moderately food insecure into marginally food insecure or food secure to marginally food insecure).
- Close match lower bound = 39% (this is the percentage of households that have been misclassified into a lower food insecurity category by the CARI using the income indicator e.g., marginally food insecure to moderate food insecure).
- Bad match = 42% (this is the percentage of households that have been badly match, i.e., classified into at least two categories above or below their food security status).

The analysis further looks at how accurately the CARI calculated using the income source and

income indicator compares CARI using FES in classifying food insecurity (vulnerable households) as a group.

Based on the classification of food insecure and food secure in Table 19, the sensitivity (error of exclusion) is 90% and specificity (error of inclusion) is 77% respectively.

Table 19: Cross Tabulation of Acute Food Insecurity (CAR_FES vs. CARI_Income)

	Food insecure with FES		Food secure with FES		Total	
	Count	Percent	Count	Percent	Count	Percent
Food insecure with Income	805	97.58	20	2.42	825	100.00
Food secure with Income	239	56.90	181	43.10	420	100.00
Total	1044	83.86	201	16.14	1245	100.00

5 Regression analysis

In this section of the analysis, regression analysis is conducted to determine the role of phone ownership and modality of interview administration in explaining the food security of the household controlling for individual and household level characteristics.

The simple regression model specified is given as follows:

$$y_i = m_i + p_i + x_i + u$$

where y_i is a binary indicator of the food security status of the ith household taking the value of one if the respondent is food secure and zero otherwise (calculated independently for FES and income indicators respectively), x_i captures the respondent's and household characteristics such as age of respondent, sex, education in years, household size, presence of disable persons in the household and dependency ratio, m_i is a binary indicator $m_i = 0/1$ taking the value of 1 if the mode of interview was F2F and zero if phone, p_i is a phone ownership indicator of the respondent $p_i = 1/0$ takes the value of one if the respondent owned a phone zero otherwise and u_i is the error term clustered at ADMIN5 level.

The regresson equation is implemented as a simple logistic regression for the whole sample. The results of the regression analysis are presented in Table 20. The regression is implemented stepwise in three different ways based on the number covariates included at a time. First, the base model which includes only the interview modality and household ownership of phone (columns 1 & 4). The second model includes the base model and the household demographics collected during the listing (columns 2 & 5). The third and final model includes all the variables while controlling for the form length (columns 3 & 6).

The results of the base model (columns 1 & 3) show that, owning a phone significantly reduces the probability of being food insecure by 19 percentage points (FES) and 31 percentage points (income choice & income change) respectively. This effect is however reduced by 5 percentage (FES) and 10 percentage (Income choice & income change) points as demographic and form length are controlled in the second and third models respectively.

Modality of interview and length of forms however are found not to have any statistical significance in explaining food insecurity classification.

In terms of the household demographics, respondent being the head of the household (7%), marital status of the household head (12%), years of education of the household head (1%) and having a member of the household with disability/chronic diseases (7% - 11%) are the significant factors that explain the food security status of the household. All these factors are found to be negatively associated with being food secure.

The analysis is also performed excluding households that received mobile phones from the research

team. This level of analysis would represent a typical CATI survey (i.e., only households with access to mobile phones are included in the sampling frame) with the conventional F2F survey sample. The results are included as Table 24 in the Annex. Although the magnitude of effect is different, the results are similar to what has been reported in Table 20.

Table 20: Logit Regression on Food Security based Outcomes

	Food Insecure with FES			Food Insecure with Income		
	(1)	(2)	(3)	(4)	(5)	(6)
Household owns phone	-0.19***	-0.14***	-0.14***	-0.31***	-0.21***	-0.21***
	(0.02)	(0.02)	(0.02)	(0.05)	(0.04)	(0.04)
Remote Modality	-0.02	-0.01	-0.01	0.04	0.06	0.06
	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)	(0.05)
List respondent age		0.00	0.00		0.00	0.00
		(0.00)	(0.00)		(0.00)	(0.00)
Household head is the respondent		0.02	0.02		-0.07*	-0.07*
		(0.02)	(0.02)		(0.04)	(0.04)
Female Household Head		0.01	0.01		-0.00	-0.00
		(0.03)	(0.03)		(0.04)	(0.04)
Married Household Head		-0.02	-0.02		-0.12**	-0.12***
		(0.03)	(0.03)		(0.04)	(0.04)
Household head years of education		-0.00*	-0.00*		-0.01	-0.01
		(0.00)	(0.00)		(0.00)	(0.00)
Household size		0.00	0.00		-0.01	-0.01
		(0.00)	(0.00)		(0.01)	(0.01)
Number of household member younger than 15		-0.01	-0.01		-0.01	-0.01
		(0.01)	(0.01)		(0.01)	(0.01)
Number of household member older than 60		0.01	0.01		0.02	0.02
		(0.02)	(0.02)		(0.04)	(0.04)
Has household member with disability/chronic disease		-0.07***	-0.07***		-0.11**	-0.10**
		(0.02)	(0.02)		(0.04)	(0.04)
Household Crowding Index		0.00	0.00		0.01	0.01
		(0.01)	(0.01)		(0.01)	(0.01)
Household has impoverished roof		0.04*	0.04*		0.01	0.01
		(0.02)	(0.02)		(0.05)	(0.05)
Household has impoverished floor		-0.00	-0.00		-0.02	-0.01
		(0.04)	(0.04)		(0.05)	(0.05)
Short Form			-0.01			-0.02
			(0.02)			(0.04)
Obs.	1245	1245	1245	1245	1245	1245

Note: Estimates are marginal effects using logit model regression. Robust standard errors are clustered at village level (Admin5) and reported in parentheses. * denotes statistical significance at 0.05 ** at 0.01 and *** at 0.001.

6 Summary & conclusions

The conclusions based on this study should be read and interpreted considering the following limitations. First, the final remote sample that has been interviewed is relatively small. Second, although there is no statistical difference between the respondents who received phones from the project and phone owners, their response may be influenced by receipt of the mobile phones. Third, the results have not been weighted to adjust for the different response rates.

The following are some key highlights from the results of the analysis from the CAR pilot study:

- The low phone ownership (25%) and mobile coverage influenced greatly the study implementation, sampling and number of households reached over the phone (remote sample). The response rate was generally low in the remote modality even though phones were distributed to potential respondents.
- Respondents over the phone generally tend to overreport their food security status compared to F2F interviews.
- The results of the food insecurity aggregation show that, rCARI-Inc methodology significantly underestimates acute food insecurity (AFI) by between 9 and 21 percentage points.
- While there is no significant difference between CARI-FES and rCARI-FES in the phone owner sub-sample, rCARI-Inc significantly underestimates food insecurity by up to 7 percentage points.
- In terms of the food security indicators, households interviewed over the phone (remote modality) reports higher food consumption (FCS), worse coping situations (rCSI and LCS) compared to when interviewed in F2F. Also, when interviewed remotely, households reported significantly higher FES compared to F2F.
- The analysis of the long- and short- LCS, FES and FCS modules (F2F) overall does not show any statistically significant difference. This means that by fine-tuning the short modules could potentially replace the longer versions in F2F surveys to reduce survey fatigue both for the interviewee and interviewer.
- The results comparing the standard F2F sub-sample with the phone owners' sub-sample (i.e., a typical CATI sample) show that using the standard CATI in regions/countries/areas where phone penetration is low tends to exclude more vulnerable and food insecure households in so doing underreport AFI.

6.1 Study implementation challenges in CAR

Although during the field work, extensive efforts were made to reduce and manage implementation challenges, a number of these remained and affected the successful implementation of the survey. These include:

1. Restricted access to the field because of security concerns.
2. Poor mobile phone network/infrastructure
    a. Most parts of the country do not have mobile phone network coverage.
    b. In places where mobile coverage exists, poor infrastructure and frequent power cuts affected call quality.
    c. Inadequate access to electricity for charging phones.
3. Low phone ownership in the population
    a. Only 25% of households in the sample owned phones.
4. Logistical challenges.
    a. Acquisition and distribution of mobile phones/SIM cards to study sample without phones.
    b. Had to rely on UNHAS (United Nations Humanitarian Air Service) for enumerator deployment affecting timely start of the survey.
5. Low response rate in the remote arm (high attrition for the remote sample).
6. Lack of an established platform for remote data collection.
    a. This affected the ability to record phone interviews.
    b. Reduced the ability of the team to perform real time monitoring of the remote data collection.
    c. Overall monitoring was affected
7. Weather disruptions/rainy season

6.2 Next steps...

1. Conduct further in-depth analysis of the data - exploring the association between the indicators especially FES, income indicators and other statistical diagnostic analysis.
2. Prepare the draft manuscripts for peer review publication.

7 References

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8 Annexes

8.1 Final survey sample size & and phone access by ADMIN Area

Table 21: Listed Sample: ADMIN Area Selection and Distribution of Phone Ownership

	Household owns phone		
	No	Yes	Total
N	1,452 (69.8%)	628 (30.2%)	2,080 (100.0%)
Geographic Region: Admin 1			
Mbomou	260 (17.9%)	425 (67.7%)	685 (32.9%)
Nana-Gribizi	1,192 (82.1%)	203 (32.3%)	1,395 (67.1%)
Geographic Region: Admin 2			
Bangassou	260 (17.9%)	425 (67.7%)	685 (32.9%)
Kaga-Bandaro	1,192 (82.1%)	203 (32.3%)	1,395 (67.1%)
Geographic Region: Admin 3			
Nana	467 (32.2%)	51 (8.1%)	518 (24.9%)
Botto	356 (24.5%)	18 (2.9%)	374 (18.0%)
Kaga-Bandoro	369 (25.4%)	134 (21.3%)	503 (24.2%)
Bangassou	260 (17.9%)	425 (67.7%)	685 (32.9%)
Geographic Region: Admin 5			
DAMAKA 2	57 (3.9%)	0 (0.0%)	57 (2.7%)
BISSEKEBOU (1,2,3 et	61 (4.2%)	3 (0.5%)	64 (3.1%)
KIA 3	97 (6.7%)	3 (0.5%)	100 (4.8%)
OUANDAGO 2	83 (5.7%)	15 (2.4%)	98 (4.7%)
YAFARA - GOUVA	77 (5.3%)	23 (3.7%)	100 (4.8%)
KONVI 1	92 (6.3%)	7 (1.1%)	99 (4.8%)
ZEFIO	92 (6.3%)	4 (0.6%)	96 (4.6%)
BEGUEDE 2	98 (6.7%)	2 (0.3%)	100 (4.8%)
BOLAFEI	78 (5.4%)	0 (0.0%)	78 (3.8%)
KPOKPO 1	88 (6.1%)	12 (1.9%)	100 (4.8%)
MISSION CATHOLIQUE	30 (2.1%)	67 (10.7%)	97 (4.7%)
MBROUA 2	70 (4.8%)	34 (5.4%)	104 (5.0%)
GRIBINGUI 1	103 (7.1%)	8 (1.3%)	111 (5.3%)
KPOPO 2	88 (6.1%)	12 (1.9%)	100 (4.8%)
GRIBINGUI 2	78 (5.4%)	13 (2.1%)	91 (4.4%)
KOBO	25 (1.7%)	54 (8.6%)	79 (3.8%)
GBIAKOTA	22 (1.5%)	44 (7.0%)	66 (3.2%)
MALIKO 2	30 (2.1%)	45 (7.2%)	75 (3.6%)
DAKPOUTOU	30 (2.1%)	50 (8.0%)	80 (3.8%)
AOUYA	17 (1.2%)	53 (8.4%)	70 (3.4%)
BEREME	21 (1.4%)	7 (1.1%)	28 (1.3%)
KEMBE-LIGNE	34 (2.3%)	31 (4.9%)	65 (3.1%)
SAOZANGANDOU	29 (2.0%)	43 (6.8%)	72 (3.5%)
LIOUANGO-TIRAILLEUR	21 (1.4%)	60 (9.6%)	81 (3.9%)
SAMBA-GAMADE	31 (2.1%)	38 (6.1%)	69 (3.3%)

8.2 F2F Attrition

Table 22: Balance Table for Randomized Questionnaire Length Assignment in F2F Survey: Completed vs. Attrition

	Own Phone			No Phone			Total F2F		
Variable	Completed	Attrition	Diff	Completed	Attrition	Diff	Completed	Attrition	Diff
List respondent age	36.27	32.65	-3.61	39.47	38.89	-0.59	38.50	37.05	-1.46
	(12.55)	(10.44)	(0.09)	(13.82)	(15.33)	(0.77)	(13.52)	(14.30)	(0.36)
Household head is the respondent	0.72	0.62	-0.11	0.79	0.68	-0.11	0.77	0.66	-0.11*
	(0.45)	(0.50)	(0.27)	(0.41)	(0.47)	(0.08)	(0.42)	(0.48)	(0.04)
Female Household Head	0.19	0.27	0.08	0.30	0.31	0.01	0.26	0.30	0.03
	(0.39)	(0.45)	(0.38)	(0.46)	(0.46)	(0.87)	(0.44)	(0.46)	(0.54)
Married Household Head	0.27	0.12	-0.15*	0.14	0.10	-0.05	0.18	0.10	-0.08*
	(0.44)	(0.33)	(0.03)	(0.35)	(0.30)	(0.24)	(0.38)	(0.30)	(0.02)
Household head years of education	7.55	7.00	-0.55	3.41	3.16	-0.25	4.66	4.30	-0.37
	(4.13)	(3.77)	(0.48)	(3.25)	(3.52)	(0.59)	(4.02)	(3.98)	(0.40)
Displaced household	0.09	0.27	0.18*	0.03	0.06	0.03	0.05	0.12	0.08*
	(0.29)	(0.45)	(0.05)	(0.17)	(0.25)	(0.30)	(0.22)	(0.33)	(0.04)
Household size	7.72	7.23	-0.49	6.08	6.23	0.15	6.57	6.52	-0.05
	(3.44)	(4.18)	(0.56)	(2.67)	(2.58)	(0.66)	(3.02)	(3.15)	(0.88)
Number of household member younger than 15	4.04	3.77	-0.27	3.23	2.98	-0.25	3.48	3.22	-0.26
	(2.27)	(2.39)	(0.57)	(1.95)	(1.49)	(0.22)	(2.08)	(1.82)	(0.20)
Number of household member older than 60	0.21	0.19	-0.02	0.17	0.18	0.00	0.19	0.18	-0.00
	(0.49)	(0.40)	(0.82)	(0.44)	(0.39)	(0.96)	(0.46)	(0.39)	(0.93)
Has household member with disability/chronic disease	0.30	0.27	-0.03	0.15	0.19	0.05	0.19	0.22	0.02
	(0.46)	(0.45)	(0.77)	(0.35)	(0.40)	(0.37)	(0.39)	(0.41)	(0.60)
Number of rooms	2.89	2.27	-0.62*	1.99	1.66	-0.33*	2.26	1.84	-0.42***
	(1.24)	(1.22)	(0.01)	(1.08)	(1.05)	(0.02)	(1.20)	(1.13)	(0.00)
Household Crowding Index	3.02	3.77	0.74	3.93	4.97	1.04*	3.66	4.62	0.96**
	(1.70)	(2.62)	(0.15)	(2.87)	(3.50)	(0.02)	(2.61)	(3.30)	(0.01)
Household has impoverished roof	0.68	0.54	-0.14	0.85	0.84	-0.01	0.80	0.75	-0.05
	(0.47)	(0.51)	(0.17)	(0.36)	(0.37)	(0.86)	(0.40)	(0.44)	(0.34)
Household has impoverished floor	0.90	0.77	-0.13	0.96	0.95	-0.01	0.94	0.90	-0.05
	(0.30)	(0.43)	(0.13)	(0.19)	(0.22)	(0.69)	(0.23)	(0.30)	(0.17)
Observations	294	26	320	675	62	737	969	88	1,057

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

8.3 Remote Attrition

Table 23: Balance Table for Randomized Questionnaire Length Assignment in Remote Survey: Completed vs. Attrition

	Own Phone			No Phone			Total Remote		
Variable	Completed	Attrition	Diff	Completed	Attrition	Diff	Completed	Attrition	Diff
List respondent age	39.03	36.09	-2.95*	39.76	39.27	-0.49	39.34	38.62	-0.72
	(12.38)	(13.01)	(0.04)	(13.94)	(14.31)	(0.73)	(13.05)	(14.11)	(0.44)
Household head is the respondent	0.78	0.68	-0.09	0.82	0.79	-0.03	0.80	0.77	-0.03
	(0.42)	(0.47)	(0.06)	(0.38)	(0.41)	(0.37)	(0.40)	(0.42)	(0.30)
Female Household Head	0.25	0.21	-0.05	0.32	0.31	-0.01	0.28	0.29	0.00
	(0.44)	(0.41)	(0.30)	(0.47)	(0.46)	(0.77)	(0.45)	(0.45)	(0.94)
Married Household Head	0.27	0.25	-0.03	0.14	0.10	-0.04	0.22	0.13	-0.08**
	(0.45)	(0.43)	(0.56)	(0.35)	(0.31)	(0.26)	(0.41)	(0.34)	(0.00)
Household head years of education	7.92	7.03	-0.89	3.54	3.26	-0.28	6.03	4.02	-2.01***
	(4.69)	(4.26)	(0.08)	(3.49)	(3.17)	(0.42)	(4.73)	(3.74)	(0.00)
Displaced household	0.15	0.08	-0.07*	0.03	0.03	-0.00	0.10	0.04	-0.06**
	(0.36)	(0.27)	(0.04)	(0.18)	(0.18)	(0.92)	(0.30)	(0.20)	(0.00)
Household size	8.39	7.50	-0.89	5.99	5.96	-0.04	7.36	6.27	-1.09***
	(3.40)	(4.42)	(0.05)	(3.05)	(2.57)	(0.91)	(3.46)	(3.10)	(0.00)
Number of household member younger than 15	4.31	3.68	-0.63**	3.13	3.13	-0.00	3.80	3.24	-0.56***
	(2.03)	(2.17)	(0.01)	(2.29)	(1.81)	(1.00)	(2.22)	(1.90)	(0.00)
Number of household member older than 60	0.21	0.24	0.03	0.13	0.19	0.05	0.18	0.20	0.02
	(0.44)	(0.64)	(0.65)	(0.41)	(0.49)	(0.21)	(0.43)	(0.53)	(0.52)
Has household member with disability/chronic disease	0.31	0.28	-0.02	0.15	0.17	0.01	0.24	0.19	-0.05
	(0.46)	(0.45)	(0.69)	(0.36)	(0.37)	(0.68)	(0.43)	(0.39)	(0.10)
Number of rooms	3.13	2.85	-0.29	1.95	1.91	-0.04	2.62	2.10	-0.53***
	(1.27)	(1.40)	(0.06)	(1.09)	(1.09)	(0.69)	(1.33)	(1.22)	(0.00)
Household Crowding Index	2.97	3.04	0.06	3.80	4.04	0.25	3.33	3.84	0.51**
	(1.26)	(1.97)	(0.74)	(2.68)	(3.02)	(0.37)	(2.04)	(2.87)	(0.00)
Household has impoverished roof	0.63	0.68	0.05	0.86	0.86	0.00	0.73	0.82	0.09**
	(0.48)	(0.47)	(0.34)	(0.35)	(0.35)	(0.99)	(0.45)	(0.38)	(0.00)
Household has impoverished floor	0.92	0.90	-0.02	0.97	0.96	-0.01	0.94	0.95	0.01
	(0.28)	(0.30)	(0.62)	(0.16)	(0.19)	(0.42)	(0.23)	(0.22)	(0.66)
Observations	157	151	308	119	596	715	276	747	1,023

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

8.4 Regression table for a typical CATI sample with F2F

Table 24: Logit Regression on Food Security based Outcomes on CATI Sample

	Food Insecure with FES			Food Insecure with Income		
	(1)	(2)	(3)	(4)	(5)	(6)
Household owns phone	-0.21***	-0.15***	-0.15***	-0.29***	-0.19***	-0.19***
	(0.03)	(0.02)	(0.02)	(0.04)	(0.04)	(0.04)
Remote Modality	0.01	0.02	0.02	0.00	0.02	0.03
	(0.04)	(0.04)	(0.04)	(0.05)	(0.06)	(0.06)
List respondent age		0.00	0.00		0.00	0.00
		(0.00)	(0.00)		(0.00)	(0.00)
Household head is the respondent		0.02	0.02		-0.07*	-0.07*
		(0.02)	(0.02)		(0.03)	(0.03)
Female Household Head		0.00	0.00		-0.02	-0.02
		(0.02)	(0.02)		(0.04)	(0.04)
Married Household Head		-0.02	-0.02		-0.13***	-0.13***
		(0.03)	(0.03)		(0.04)	(0.04)
Household head years of education		-0.00*	-0.01*		-0.01*	-0.01*
		(0.00)	(0.00)		(0.00)	(0.00)
Household size		0.00	0.00		-0.01	-0.01
		(0.00)	(0.00)		(0.01)	(0.01)
Number of household member younger than 15		-0.01	-0.01		-0.02	-0.02
		(0.01)	(0.01)		(0.01)	(0.01)
Number of household member older than 60		0.02	0.02		0.04	0.04
		(0.02)	(0.02)		(0.04)	(0.04)
Has household member with disability/chronic disease		-0.08***	-0.08***		-0.14**	-0.14**
		(0.02)	(0.02)		(0.05)	(0.05)
Household Crowding Index		0.00	0.00		0.01	0.01
		(0.01)	(0.01)		(0.01)	(0.01)
Household has impoverished roof		0.05**	0.05**		0.02	0.03
		(0.02)	(0.02)		(0.06)	(0.06)
Household has impoverished floor		-0.01	-0.00		-0.01	-0.01
		(0.04)	(0.04)		(0.06)	(0.06)
Short Form			-0.03			-0.06
			(0.02)			(0.04)
Obs.	1126	1126	1126	1126	1126	1126

Note: CATI Sample includes all F2F respomdents and Remote phone-owners, removing 119 Remote surveys without phone. Estimates are marginal effects using logit model regression. Robust standard errors are clustered at village level (Admin5) and reported in parentheses. * denotes statistical significance at 0.05 ** at 0.01 and *** at 0.001.

8.5 Supporting documents

8.5.1 CARI resources

- CARI methodology Consolidated Approach for Reporting Indicators of Food Security (CARI) Guidelines - Third edition
- FCS
- rCSI
- LCS
- FES

8.5.2 Listing questionnaire

Formulaire de liste des ménages rCARI

Welcome to WFP form for HH listing

Date of the listing

2024-01-10

2024-01-10

ADMIN1Name

- Mbomou
- Nana-Gribizi

ADMIN2Name

ADMIN3Name

ADMIN5Name

ADMIN5Name confirmation

- Yes
- No

Village if different from list

Enumerator Name

Geographic Coordinates

ENUMERATOR: try for at least 60 seconds to get the exact location coordinates of the house.

latitude (x.y °)

longitude (x.y °)

altitude (m)

accuracy (m)

What is the house number/address?

Enter the house number if applicable, otherwise skip

Description of the house

Describe some unique features that identifies the house

Is the house/structure permanent structure?

ENUMERATOR: Assess if the building is temporary like a tent, hut etc or a permanent building

- Yes
- No

Hello, my name is , and I work with the World Food Programme [WFP]/ICASEES. We are conducting a survey in the Central African Republic to validate food security indicators to help in programing decisions. All the information you provide will remain strictly confidential and will not be communicated to others such as government agencies or authorities. We are listing potentail participants/households/respondents who may be contacted in the coming days to be interviewed either in person or through phone calls. Will you agree to be contacted in again for the full interview either by phone or in-person visit starting from 25/07/2023?

ENUMERATOR: Take time to read this statement in full to the respondent. Explain if necessary to clarify any concerns of the respondent such as fear of victimization, security or mistrust. Assure the respondent the information would be confidential.

Do you agree to participate?

- Yes
- No

Please indicate the reason for refusal

Select the most important reason for refusal.

- I am not interested
- I won't be available in the territory at the time
- For security reasons
- For personal reasons I cant share
- Due to bad previous experience
- No reason

Select the sex of the respondent

Observe and select the gender the respondent

- Female
- Male

Select the age bracket of the respondent

Select the age bracket that the respondent may be classified. You may ask if possible else make a comfortable guess

- < 18 years
- 18-25 years
- 25-35 years
- 45-55 years
- >55 years

What is your full name?

_____

How old are you?

_____

What is 's relationship to the household head?

- Household head (self)
- Spouse
- Child
- Parent
- Sibling
- Uncle/Aunt
- Non-family relation

What is the name of the household head?

Please enter the full name of the household head as he/she is known by in the village

_____

Status of the household

- Permanent resident
- Displaced
- Refugee

Sex of the household head?

- Female
- Male

Is the household head married?

- Yes
- No

Years of education of the household head

_____

How many people are in this household?

_____

How many members in your household are younger than 15years of age?

_____

How many members in your household are older than 60years of age?

_____

Is there anyone with disability/chronic disease in your household?
-

- Yes
- No

How many rooms are in the house?
by rooms, ENUMERATOR clarify that you mean bedrooms

_____

Does the household have access to mobile phone?
It could the head of the household or anyone in the household

- Yes
- No

What is the primary phone number for household head or which you may be reached?

_____

Is there a second phone number you may be reached on?

- Yes
- No

What is the secondary phone number for household head?

Demandez au répondant s'il existe un autre numéro de téléphone auquel il peut être joint si le premier n'est pas accessible.

_____

What is the third phone number for household head?

_____

Please select the option that best describes the roofing material of this house

FOR THE ENUMERATOR: Answer this by observing the roof yourself [Only ask the respondent if you are not sure of the material].

- No roof
- Thatch/Palm leaf
- Sod
- Rustic mat
- Palm/bamboo
- Wood planks
- Cardboard
- Metal/tin
- Wood planks
- Calamine/Cement fibre
- Ceramic tiles
- Cement
- Roofing shingles
- Don't know
- Other (specify)

Please select the option that best describes the floor material of this house

ENUMERATORS: Only ask this if you are not sure after observing the floor yourself

- ◯ Earth/sand
- ◯ Dung
- ◯ Wood planks
- ◯ Palm/bamboo
- ◯ Parquet or polished wood
- ◯ Vinyl or asphalt strips
- ◯ Ceramic tiles
- ◯ Cement
- ◯ Carpet
- ◯ Other (specify)

The team may contact you in the coming days, what time(s) of the day would be convenient for us to visit or contact you by phone?

Ask the respondent to provide at least 3 different times that they would be available

- ☐ 8am-11pm
- ☐ 12pm - 2pm
- ☐ 4pm - 6pm
- ☐ after 7pm
- ☐ Anytime in the day
- ☐ Only in the evening

May I take a photo of the house/any landmark close to you building, to help the interviewer who returns to conduct an interview can find you?

- ◯ Yes
- ◯ No

Image of house/landmark structure to identify the household.

Please ensure the picture taken shows distinctive features such that the house can be identified when the field team visits the village.

Click here to upload file. (< 5MB)

Image of house/landmark structure to identify the household.

Please ensure the picture taken shows distinctive features such that the house can be identified when the field team visits the village.

Click here to upload file. (< 5MB)

ENUMERATOR: Check with your phone whether there is coverage in the house location or not

Double check if the household can receive mobile calls

- ◯ Yes
- ◯ No

Thank you for your time. The team may contact you or your household for the full interview later.